

Ain Shams University

Faculty of Computer and Information Sciences (FCIS)

Task manager

Academic Year: 2025 / 2026

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1. Project Objective

The primary objective of the **Student Task & Study Planner** is to provide Android users—specifically students—with a lightweight, intuitive, and efficient local management tool to track academic tasks, assignments, and study schedules.

Key objectives include:

- **Persistent Local Storage:** Leveraging an embedded SQLite database (`SQLiteOpenHelper`) to ensure data remains persistent, reliable, and accessible offline without requiring network dependency.
- **Structured UI & UX:** Offering clean menu-driven and list-based interfaces to navigate seamlessly between creating, reviewing, updating, and analyzing study tasks.
- **Data-Driven Insights:** Providing users with statistical breakdowns of task progress (Completed, In Progress, Not Started) to foster better time management habits.

2. Project Functionalities

The application provides end-to-end CRUD (Create, Read, Update, Delete) capabilities alongside analytical summary views:

A. Navigation & Dashboard (`MainActivity`)

- **Centralized Navigation Hub:** Provides direct access to Add Task, View Tasks, and Statistics interfaces via styled buttons and an Options Menu (`main_menu.xml`).
- **About / Information:** Includes an Options Menu item displaying application metadata via Android `Toast`.

B. Task Creation (`AddTaskActivity`)

- **Form Validation:** Validates required input fields (e.g., ensuring title fields are non-empty before saving).
- **Attribute Management:** Allows input of title, detailed description, subject title, priority selection via `Spinner` (Low, Medium, High), and completion status via `RadioGroup` (Not Started, In Progress, Completed).
- **Database Insertion:** Inserts newly formed `Task` entities directly into the SQLite database table (`TABLE_TASKS`).

C. Task Management & RecyclerView (`TaskListActivity` & `TaskAdapter`)

- **Dynamic Task Listing:** Displays stored tasks inside a dynamic `RecyclerView` using custom layouts (`task_row.xml`) with styled grid-like borders.

- **Interactive Menus:**
 - **Normal Click:** Opens full task details for editing.
 - **Long Click (Context Menu):** Triggers an Android Context Menu allowing fast Edit and Delete actions.
 - **Row Menu (Popup Menu):** Triggers an overflow menu (three-dot menu) per list item to perform Edit or direct SQLite Deletion with auto-refreshing UI.

D. Task Updating & Deletion (`TaskDetailsActivity`)

- **Pre-populated Fields:** Fetches existing task records using `task_id` extras passed through Android `Intent` and fills input controls dynamically.
- **Database Update & Delete Operations:** Allows users to modify details or remove records from SQLite with feedback provided via `Toast` notifications.

E. Real-time Progress Tracking (`StatisticsActivity`)

- **Task Analytics:** Dynamically queries the SQLite database to compute and display total task counts and categorizes them by status (*Completed*, *In Progress*, and *Not Started*).

Screenshot 1: Main Dashboard (`MainActivity`)



Screenshot 2: Add Task Form (AddTaskActivity)

ADD NEW TASK

Task Title

Enter task title

Description

Enter task description

Subject

Enter subject

Priority

Low

Status

☒ Not Started

☐ In Progress

☐ Completed

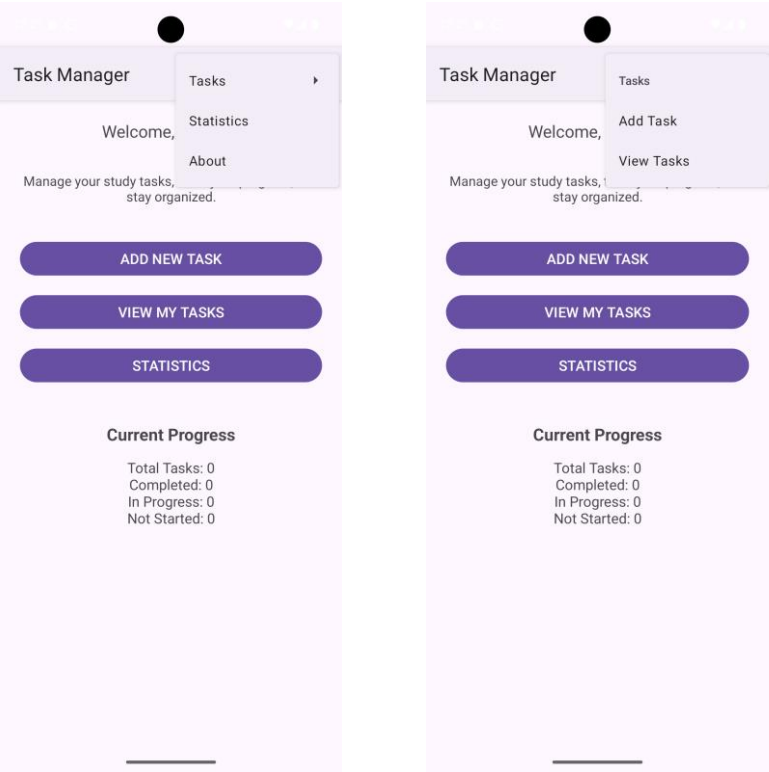
SAVE TASK

Screenshot 3: Task List View (TaskListActivity)

12:25 MY

TITLE	DESCRIPTION	S
Handle Popup Menu	Popup menu with all user options	P

Screenshot 4: Sub & Popup Menus



Screenshot 5: Statistics Screen (StatisticsActivity)

